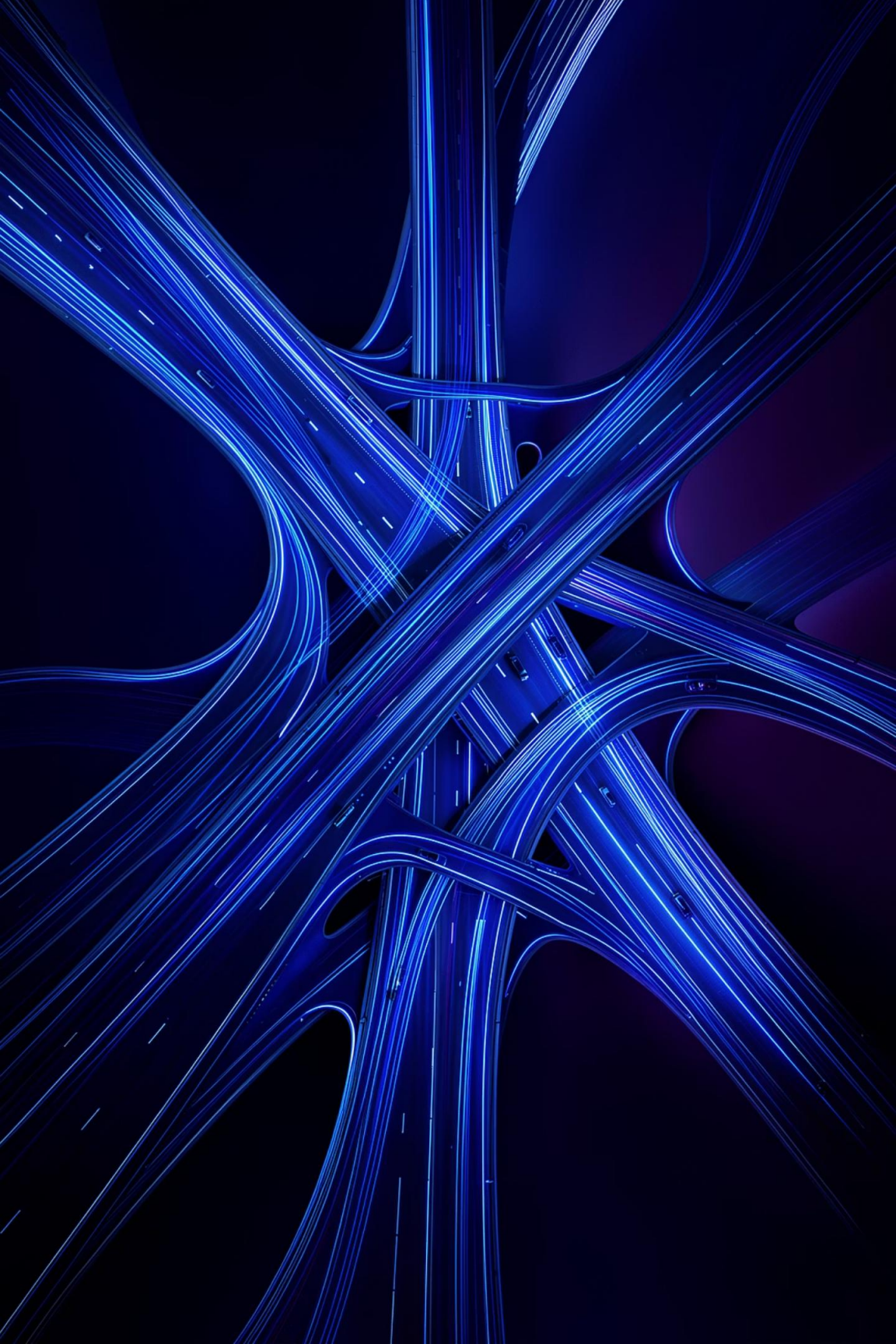




Accident-Prone Area Prediction using Machine Learning

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Problem Statement

Can machine learning predict *where* accidents are most likely to occur — before they happen?

Road traffic accidents remain a leading cause of fatalities and injuries worldwide. Traditional accident analysis relies on post-incident reporting, which is reactive rather than preventive. There is a critical need for a **data-driven, proactive system** that identifies accident-prone zones using historical data, environmental factors, and ML models — enabling timely intervention by authorities.



Objectives

1

Predict Risk Zones

Identify geographic locations with high accident probability using ML classification and regression models.

2

Analyse Contributing Factors

Study the influence of road type, weather, time of day, and traffic density on accident occurrence.

3

Visualise Hotspots

Generate intuitive heatmaps and geospatial visualisations of predicted accident-prone corridors.

4

Support Decision-Making

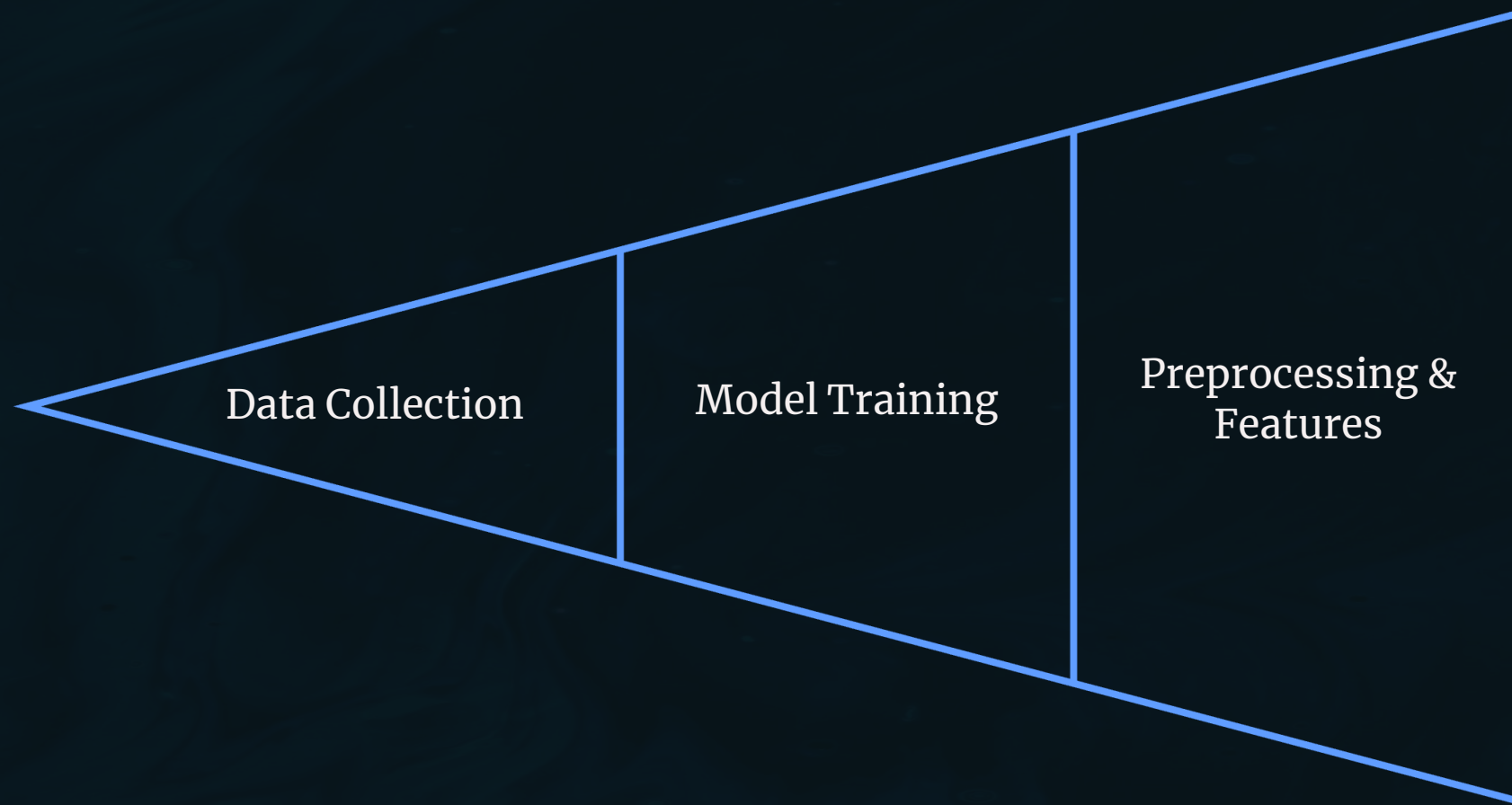
Provide actionable insights to urban planners, traffic police, and road safety authorities.

Literature Survey

Reference	Methodology	Key Finding
Ahmed et al. (2019)	Random Forest on US accident data	Weather and road surface are top predictors of severity
Parsa et al. (2020)	XGBoost with spatial clustering	Hotspot detection accuracy improved by 18% over baseline
Ito & Sugiyama (2021)	LSTM for time-series accident trends	Temporal patterns significantly enhance prediction recall
Moosavi et al. (2022)	Deep learning + GIS mapping	Geo-contextual features are critical for zone-level accuracy

❗ Existing works lack integration of real-time traffic and weather feeds — a gap this project addresses.

Proposed Solution Overview



The system ingests multi-source data, engineers spatial and temporal features, trains ensemble models, and outputs interactive hotspot maps for real-world deployment.

Input

Historical accident records, weather API, road metadata

Model

Random Forest, XGBoost, SVM ensemble

Output

Risk-level heatmap with severity classification

Dataset Details

Primary Dataset

LI Accidents in Arizona — 2.8 million records across 49 states (2016–2021). Features include GPS coordinates, severity (1–4), weather conditions, road features, and time-of-day.

- 49 attributes per record
- Target variable: Accident severity (1–4)
- Geospatial: Latitude / Longitude

Weather Features

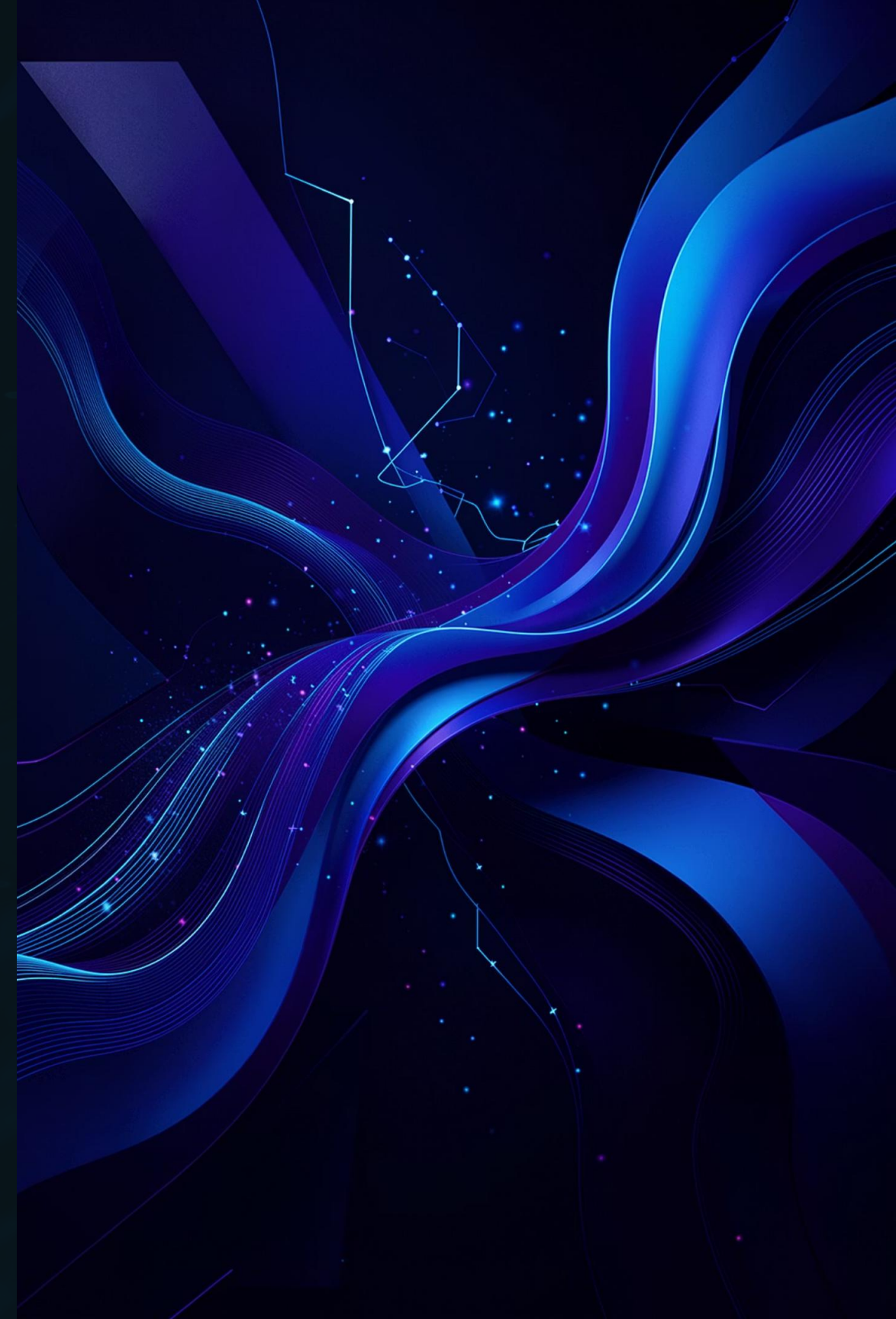
Temperature, visibility, wind speed, precipitation

Road Features

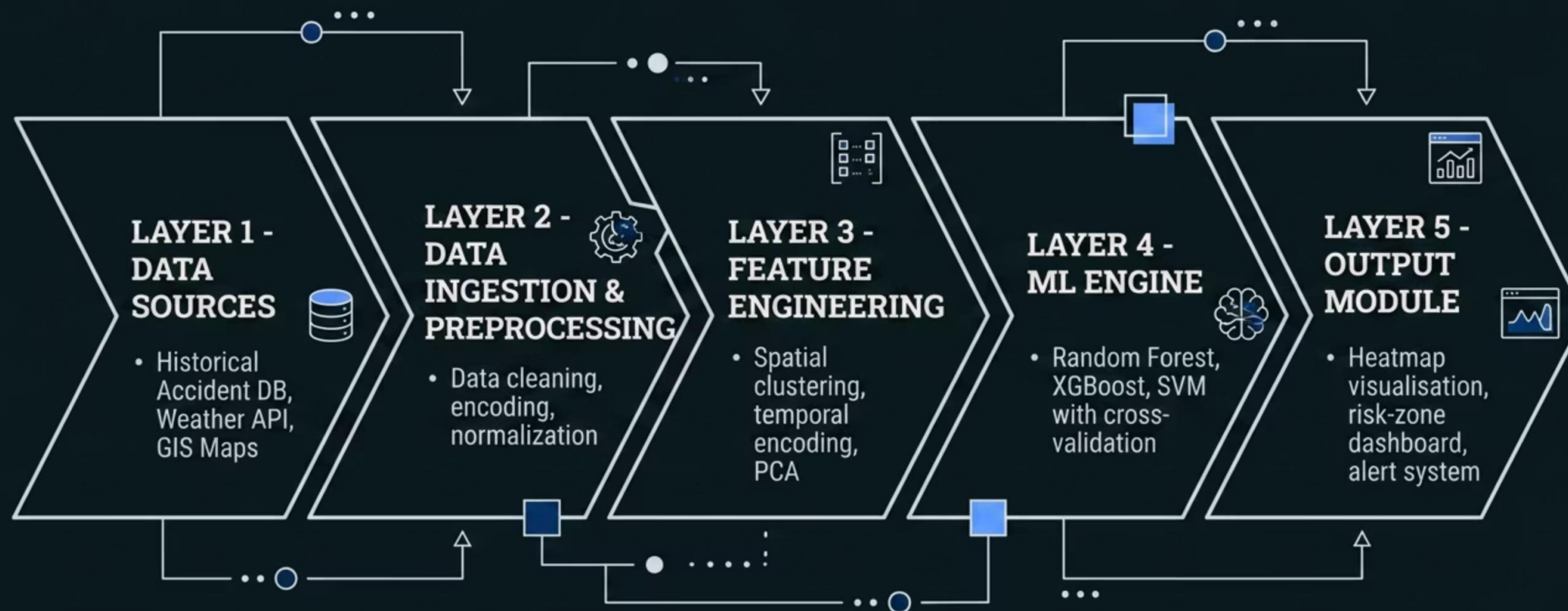
Junction, traffic signal, crossing, roundabout

Temporal Features

Hour, day of week, month, public holiday flag



System Architecture



The five-layer architecture ensures clean separation between data ingestion, feature engineering, model inference, and presentation — enabling modular upgrades and real-time integration.

Tools & Technologies



Python 3.10

Core programming language for data processing, model training, and automation pipelines.



Jupyter & Pandas

Exploratory data analysis, data wrangling, and feature engineering in an interactive environment.



Scikit-learn & XGBoost

Primary ML libraries for implementing Random Forest, SVM, and gradient boosting classifiers.



Flask / Streamlit

Lightweight web framework for deploying the prediction model as an interactive user-facing dashboard.



Folium / GeoPandas

Geospatial visualisation tools for generating interactive accident heatmaps on real maps.

Sample Input & Output

Sample Input

Location: 34.0522° N, 118.2437° W

Time: Friday, 08:15 AM

Weather: Rain, Visibility 0.5 mi

Road Type: Urban intersection with signal

Temperature: 16°C | Wind: 22 km/h

Predicted Output

Heatmap flags this zone . Alert dispatched to traffic management centre. Suggested action: Deploy patrol unit and activate variable speed signs.

Risk Level

HIGH — Severity 3

Confidence

87.4%

References

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